

WPU SABLAE Portal Source Code Documentation

CLASS / lms_scheduling

2026-07-27

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WPU SABL Ae Portal — Source Code Documentation

Project: lms_scheduling (Faculty Scheduling & Classroom LMS)

Stack: PHP 8+ (strict types), MySQL/MariaDB, Bootstrap 5, PDO

Entry point: index.php → login.php or dashboard.php

This documentation describes all application source code organized by functional module. Each module page lists source files, database tables, roles, and key behaviors.

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#	Module	Document
1	Core infrastructure	01-core-infrastructure.md
2	Authentication & authorization	02-authentication.md
3	Installation & configuration	03-installation.md
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6	Dean (college)	06-dean.md
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9	General Education (GEN ED)	09-gened.md
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User Roles

Role	Scope	Primary responsibilities
super_admin	Institution	Admin accounts, faculty inventory, teaching-load reports
admin	Institution	Colleges, deans, GEN ED account, conflict requests, global reports
dean	One college	Programs, chairs, faculty, rooms, courses, scheduling, auto-schedule
program_chair	One program (within college)	Faculty, courses, schedules, students, registrations
gened	Institution (GE only)	GE courses, faculty, rooms, schedules, assignments
faculty	Assigned classes	Classrooms, attendance, assessments, teaching load
student	Enrolled classes	Join classrooms, EduTools, wellness check-in

Directory Layout

CLASS/	
— api/	# JSON REST endpoints
— assets/css/	# Application stylesheets
— assets/js/	# Client-side scripts
— config/	# config.php (from config.example.php)
— docs/	# This documentation set
— includes/	# Shared PHP libraries & layout partials
— install/	# schema.sql
— uploads/	# User-uploaded files (attachments, photos)
— *.php	# Role-specific page controllers
— install.php	# Database installer
— upgrade_roles.php	# Schema migrations & role upgrades

Request Lifecycle

1. **Bootstrap** — Page includes includes/auth.php (starts session) and often includes/functions.php.
 2. **Authorization** — require_login() or require_role([...]) gates access.
 3. **Business logic** — Inline in page file or delegated to includes/*_helpers.php.
 4. **Presentation** — includes/header.php + HTML + includes/footer.php.
 5. **Navigation** — includes/admin_nav.php renders role-specific sidebar via render_role_nav_sections().
-

Database Overview

Core entities: colleges, users, programs, college_programs, faculty, courses, rooms, schedules.

Supporting: conflict_logs, conflict_requests, schedule_change_requests, dean_activity_logs, admin_activity_logs.

GE-specific: ge_course_colleges, ge_schedule_targets, faculty_specializations.

LMS: online_classrooms, classroom_students, classroom_enrollments, classroom_content, classroom_assessments, classroom_scores, classroom_submissions, classroom_attendance*.

Messaging: internal_messages.

Registration: student_registration_requests.

Full DDL: install/schema.sql.

Maintenance Scripts

File	Purpose
install.php	Initial database creation and default admin/GEN ED accounts
upgrade_roles.php	Idempotent ALTER TABLE migrations (email, messaging, wellness, etc.)
fix_room_code_scope.php	Repairs rooms.room_code_scope generated column data

Generated from source analysis of the CLASS codebase.

Exported documents

All modules combined into single files (with code appendix):

Format	File
PDF	WPU_SABLAe_Source_Code_Documentation.pdf
Word	WPU_SABLAe_Source_Code_Documentation.docx
HTML	WPU_SABLAe_Source_Code_Documentation.html
Markdown	SOURCE_CODE_DOCUMENTATION.md

Regenerate: C:\xampp\php\php.exe docs\build_exports.php

Module 1: Core Infrastructure

Shared foundation used by every page in the application.

Source Files

File	Purpose
includes/db.php	PDO singleton (db()), MySQL connection with XAMPP-friendly error messages
includes/functions.php	General utilities: DB introspection, scheduling rules, classroom helpers, room codes, conflict detection
includes/header.php	HTML <head>, Bootstrap/CSS, top navigation shell
includes/footer.php	Closing markup, script includes
includes/student_tooltip.php	Tooltip attributes for UI hints (app_tooltip_attr())
index.php	Redirects to dashboard.php (logged in) or login.php
logout.php	Clears session, records last_logout_at, redirects to login

includes/db.php

function db(): PDO

- Loads config/config.php.
 - Creates a static PDO connection to DB_HOST:DB_PORT/DB_NAME.
 - Sets ERRMODE_EXCEPTION and FETCH_ASSOC.
 - Throws a readable RuntimeException when MySQL is not running (errno 2002).
-

includes/functions.php — Key Function Groups

Database introspection

Function	Description
db_column_exists(\$table, \$column)	Checks information_schema.COLUMNS
db_table_exists(\$table)	Checks information_schema.TABLES
db_first_existing_column(\$table, \$cols, \$fallback)	Picks first column that exists (migration-safe)

Scheduling primitives

Function	Description
schedule_days_list()	Returns Mon–Sun array
parse_day_set(\$value) / days_to_set(\$days)	MySQL SET ↔ PHP array
time_to_minutes(\$time) / minutes_to_time(\$min)	Time arithmetic
intervals_overlap(\$s1,\$e1,\$s2,\$e2)	Overlap test
validate_schedule_rules(...)	Enforces TIME_MIN/MAX, MIN_CLASS_MINUTES, MIN_GAP_MINUTES, MAX_CONSECUTIVE_HOURS
checkConflicts(...)	Faculty, room, and cross-college room conflict detection
log_conflicts(\$scheduleId, \$conflicts)	Writes to conflict_logs
create_conflict_request(\$payload)	Inserts conflict_requests for admin review
sql_scope_for_college(\$alias, \$collegeId)	SQL fragment limiting rows to a college

Room management

Function	Description
<code>room_code_taken_for_college(...)</code>	Uniqueness within college scope
<code>room_code_taken_for_gened(...)</code>	Uniqueness for GE rooms
<code>next_auto_room_code_for_college(...)</code>	Auto-generates room codes
<code>next_auto_room_code_gened()</code>	GE room code generator
<code>detect_cross_college_room_conflicts(...)</code>	Shared physical room across colleges

Program / year level

Function	Description
<code>dean_program_year_levels_map(\$collegeId)</code>	Map program → year levels
<code>program_defined_year_levels(\$programId)</code>	Levels for a college_programs row
<code>program_year_levels_replace(\$programId, \$levels)</code>	Replace year levels
<code>sort_schedule_year_levels(\$levels)</code>	Natural sort (1st, 2nd, ...)

Classroom / LMS helpers

Function	Description
<code>classroom_alloc_unique_join_code()</code>	8-char join code for online_classrooms
<code>course_alloc_unique_classroom_code()</code>	Unique courses.classroom_code
<code>classroom_content_sanitize_html(\$html)</code>	Safe HTML for rich content
<code>classroom_content_store_attachment(\$file)</code>	Saves uploads under uploads/classroom/
<code>classroom_assessment_normalize_type(\$type)</code>	written_work, performance_task, assignment, quiz
<code>classroom_grade_question_submission(...)</code>	Auto-grades objective questions

General Education (GE)

Function	Description
<code>is_ge_program_scope(\$program)</code>	Detects GE program label
<code>ge_courses_offered_to_college(\$collegeId)</code>	GE courses mapped via ge_course_colleges
<code>dean_schedule_scope_sql(\$collegeId, ...)</code>	Dean sees college + applicable GE schedules
<code>weekly_schedule_gened_can_join_online_live(...)</code>	GE live session access rules

Suggestions

Function	Description
<code>suggest_available_usernames(\$preferred)</code>	Username collision avoidance
<code>suggest_available_course_codes(\$preferred, \$collegeId, \$isGened)</code>	Course code suggestions

includes/schedule_helpers.php

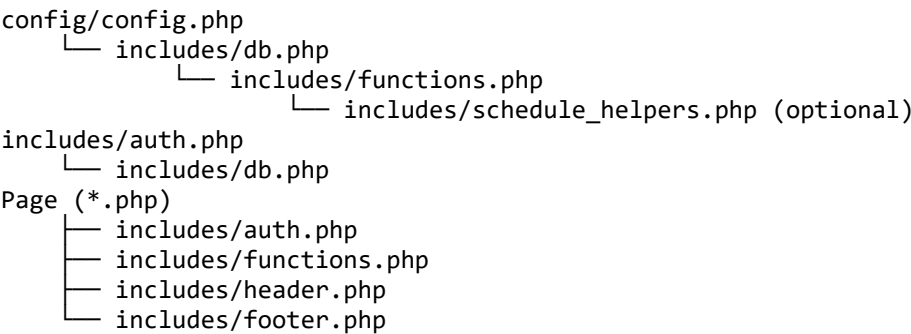
Function	Description
<code>render_schedule_form(\$defaults, \$options)</code>	Shared HTML form for add/edit schedule pages
Used by <code>add_schedule.php</code> , <code>edit_schedule.php</code> , <code>genedit_schedule.php</code> .	

Session & Config Constants

Defined in `config/config.example.php` (copy to `config/config.php`):

Constant	Purpose
<code>DB_*</code>	Database connection
<code>SESSION_NAME</code>	PHP session cookie name (<code>FSS_SESSION</code>)
<code>BASE_PATH</code>	Application root directory
<code>TIME_MIN, TIME_MAX</code>	Allowed class hours (06:00–22:00)
<code>MIN_CLASS_MINUTES, MIN_GAP_MINUTES</code>	Scheduling constraints
<code>MAX_CONSECUTIVE_HOURS</code>	Faculty consecutive teaching limit
<code>MESSAGING_THREAD_MAX_MESSAGES</code>	FIFO cap per message thread
<code>MAIL_*</code>	SMTP / credential email settings
<code>WELLNESS_AI_*</code>	Groq/Ollama wellness companion

Dependencies Between Core Files



Module 2: Authentication & Authorization

Session-based login with role-scoped access control.

Source Files

File	Purpose
<code>login.php</code>	Multi-role login UI; student self-registration
<code>login1.php</code>	Alternate/simplified login page
<code>logout.php</code>	Session destruction
<code>includes/auth.php</code>	Session helpers, role checks, ID resolution
<code>profile_photo.php</code>	Upload/remove user profile photo

File	Purpose
includes/profile_photo_helpers.php	Photo storage under uploads/profile/

Login Flow (login.php)

1. If `$_SESSION['user_id']` exists → redirect to `dashboard.php`.
 2. User selects role tab: Student, Admin, Program Chair, Dean, Faculty.
 3. **Login POST:** Validates username/password against users where `is_active = 1`.
 4. On success, populates session:
 - `user_id`, `username`, `full_name`, `role`
 - `college_id`, `assigned_program`, `admin_log_title` (when applicable)
 - Resolves `faculty_id` / `student_id` via helper functions
 - Updates `last_login_at`, `last_seen_at`
 5. **Student registration POST** (students only): Calls `submit_student_registration()` → `student_registration_requests`.
-

Session Keys

Key	Set when
<code>user_id</code>	All authenticated users
<code>username</code> , <code>full_name</code> , <code>role</code>	All authenticated users
<code>college_id</code>	Dean, program chair, college-scoped faculty
<code>assigned_program</code>	Program chair
<code>admin_log_title</code>	Admin / super_admin display title
<code>faculty_id</code>	Faculty user linked to faculty row
<code>student_id</code>	Student linked to classroom_students row

includes/auth.php — Functions

Access control

Function	Behavior
<code>require_login()</code>	Redirect to <code>login.php</code> if no session; touches <code>last_seen_at</code>
<code>require_role(array \$roles)</code>	<code>require_login()</code> + 403 if role not in list
<code>current_user()</code>	Returns session user array or null

Role predicates

Function	True when role is
<code>is_super_admin()</code>	<code>super_admin</code>
<code>is_admin()</code>	<code>admin</code>
<code>is_dean()</code>	<code>dean</code>
<code>is_program_chair()</code>	<code>program_chair</code>
<code>is_faculty()</code>	<code>faculty</code>

Function	True when role is
<code>is_gened()</code>	<code>gened</code>
<code>is_student()</code>	<code>student</code>

Scope helpers

Function	Description
<code>current_college_id()</code>	Session college ID or null
<code>current_program_scope()</code>	Program chair's assigned_program
<code>dean_college_id_or_fail()</code>	Requires dean + college; exits 403 otherwise
<code>dean_or_program_chair_college_id_or_fail()</code>	Dean or chair with college
<code>program_scope_or_fail()</code>	Program chair with assigned program
<code>college_name_by_id(\$id)</code>	Lookup from colleges

Identity resolution

Function	Description
<code>resolve_faculty_id_for_user(\$userId)</code>	Links users ↔ faculty by user_id or name match
<code>resolve_student_id_for_user(\$userId)</code>	Links users ↔ classroom_students
<code>faculty_college_id(\$facultyId)</code>	Home college for a faculty row

Activity

Function	Description
<code>auth_touch_user_presence(\$userId)</code>	Updates users.last_seen_at
<code>auth_user_activity_columns_ready()</code>	Checks migration for activity columns
<code>log_dean_activity(\$actionType, \$details)</code>	Inserts dean_activity_logs
<code>verify_admin_password(\$password)</code>	Re-auth for sensitive admin actions

Profile Photos

`includes/profile_photo_helpers.php`

Function	Description
<code>profile_photo_store(\$userId, \$file)</code>	Validates image, saves to uploads/profile/
<code>profile_photo_url(\$userId)</code>	Public URL for
<code>profile_photo_remove(\$userId)</code>	Deletes file and clears DB column

Requires users.profile_photo column (added by upgrade_roles.php).

Security Notes

- Passwords stored with `password_hash()` / verified with `password_verify()`.
- Role checks happen server-side on every protected page via `require_role()`.
- No JWT; classic PHP server sessions (SESSION_NAME cookie).

- Admin password re-verification available via `verify_admin_password()` for destructive operations.

Module 3: Installation & Configuration

Database setup, schema migrations, and environment configuration.

Source Files

File	Purpose
<code>install.php</code>	Browser-based one-time installer
<code>install/schema.sql</code>	Full database DDL
<code>upgrade_roles.php</code>	Incremental schema migrations
<code>fix_room_code_scope.php</code>	Data repair for room scope column
<code>config/config.example.php</code>	Template configuration
<code>config/config.php</code>	Live config (gitignored; copy from example)

Installation (`install.php`)

Trigger: POST or `?run=1`

Steps: 1. Connect to MySQL without database selected. 2. CREATE DATABASE IF NOT EXISTS with utf8mb4. 3. Execute all statements from `install/schema.sql` (split on `;`). 4. Seed default admin user if none exists (DEFAULT_ADMIN_* constants). 5. Seed default gened user if none exists (DEFAULT_GENED_* constants).

Output: Success message with default credentials or error details.

Schema (`install/schema.sql`)

Creates tables (InnoDB, utf8mb4):

Organization

- `colleges` — College codes, names, dean link
- `users` — All login accounts (7 roles)
- `programs` / `college_programs` — Program catalog per college
- `programs_year_levels` / `program_year_levels` / `program_sections` — Academic structure

Academic resources

- `faculty` — Instructor records, max hours, GE flag
- `courses` — Course catalog (college or GE)
- `rooms` — Physical rooms with `room_code_scope` generated uniqueness
- `schedules` — Class offerings (faculty + course + room + time)

Workflow & audit

- conflict_logs, conflict_requests
- schedule_change_requests
- dean_activity_logs, admin_activity_logs

GE extensions

- ge_course_colleges, ge_schedule_targets
- faculty_specializations

LMS & messaging

- online_classrooms, classroom_students, classroom_enrollments
- classroom_content, classroom_content_attachments
- classroom_messages, classroom_assessments, classroom_scores, classroom_submissions
- classroom_attendance_sessions, classroom_attendance_records
- internal_messages
- student_registration_requests

Migrations (upgrade_roles.php)

Idempotent migration runner using `exec_safe()` — ignores duplicate column/index errors.

Typical additions (run after initial install): - `users.email`, `last_login_at`, `last_seen_at`, `last_logout_at`, `profile_photo` - `users.role` enum extensions (`super_admin`, `gened`, `student`) - Messaging tables and memo columns - Classroom / assessment / attendance tables - Wellness-related columns - Admin activity log schema - Student registration workflow

Usage: Open in browser and POST to apply pending migrations.

Configuration (config/config.example.php)

Section	Key settings
Database	<code>DB_HOST</code> , <code>DB_PORT</code> , <code>DB_NAME</code> , <code>DB_USER</code> , <code>DB_PASS</code>
Bootstrap accounts	<code>DEFAULT_ADMIN_*</code> , <code>DEFAULT_SUPER_ADMIN_*</code> , <code>DEFAULT_GENED_*</code>
GE messaging	<code>GE_DEAN_USER_ID</code> , <code>GE_DEAN_NAME_HINT</code>
Email	<code>MAIL_ENABLED</code> , <code>MAIL_SMTP_*</code> , <code>MAIL_LOGIN_URL</code> , <code>APP_BASE_URL</code>
Scheduling rules	<code>TIME_MIN</code> , <code>TIME_MAX</code> , <code>MIN_GAP_MINUTES</code> , <code>MAX_CONSECUTIVE_HOURS</code>
Wellness AI	<code>WELLNESS_AI_ENABLED</code> , <code>WELLNESS_AI_API_KEY</code> , <code>WELLNESS_AI_MODEL</code>

Setup:

```
cp config/config.example.php config/config.php
# Edit DB credentials and URLs
# Open install.php in browser
# Run upgrade_roles.php for latest schema
```

Room Scope Repair (fix_room_code_scope.php)

Maintenance utility for `rooms.room_code_scope` — a generated column ensuring unique room codes per college or GE scope (G|code vs C|college_id|code).

Web Server

- `web.config` — IIS rewrite rules (if deployed on Windows IIS).
- `uploads/web.config` — Restricts script execution in upload directory.

Module 4: Super Admin

Highest privilege tier for institution-wide administration beyond standard admin.

Role: `super_admin`

Source Files

File	Access	Purpose
<code>super_admin_admins.php</code>	<code>super_admin</code>	Create/manage admin user accounts
<code>super_admin_faculty_inventory.php</code>	<code>super_admin</code>	Cross-college faculty roster view
<code>super_admin_teaching_load_report.php</code>	<code>super_admin</code>	Institution-wide teaching load in units
<code>settings.php</code>	All roles (incl. <code>super_admin</code>)	Profile, password, photo
<code>dashboard.php</code>	All roles	Super-admin-specific dashboard stats block

Navigation (includes/admin_nav.php)

`render_super_admin_nav_sections()` provides:

Nav item	File
Dashboard	<code>dashboard.php</code>
Manage admins	<code>super_admin_admins.php</code>
Faculty inventory	<code>super_admin_faculty_inventory.php</code>
Teaching load report	<code>super_admin_teaching_load_report.php</code>
Messages	<code>messages.php</code>
Settings	<code>settings.php</code>

`super_admin_admins.php`

- Lists all users with `role = 'admin'`.
 - CRUD: username, full name, email, active flag, optional password reset.
 - Can generate temporary password and email via `send_account_credentials_mail()`.
 - Activity logged via `log_admin_activity()`.
-

super_admin_faculty_inventory.php

- Read-only (or manage) view of faculty across all colleges.
 - Filters by college, department, employment status.
 - Links faculty rows to linked users accounts.
-

super_admin_teaching_load_report.php

- Aggregates scheduled units per faculty from schedules + courses.
 - Filters: college, semester, school year.
 - Export-friendly tabular report.
-

Dashboard Metrics (super_admin section in dashboard.php)

When role === 'super_admin', additional stats include:

- Total schedules (all time)
- Colleges / departments count
- Faculty members, student enrollments
- Active conflict alerts, resolved conflicts (30 days)
- Room double-booking risks, faculty time overlaps
- Scheduled classes current term

Auto-refreshes every 30 seconds.

Related Tables

Table	Usage
users	Admin account records (role = 'admin')
faculty	Inventory source
schedules, courses	Teaching load calculation
conflict_logs, conflict_requests	Conflict metrics
admin_activity_logs	Audit trail

Bootstrap

Default super admin credentials from config (created manually or via install extension):

- DEFAULT_SUPER_ADMIN_USERNAME / DEFAULT_SUPER_ADMIN_PASSWORD

Change password immediately after first login.

Module 5: Admin (Institution)

College-level and institution-wide management for standard administrators.

Source Files

File	Purpose
admin_colleges.php	CRUD colleges (code, name, status)
admin_deans.php	Create/manage dean accounts per college
admin_gened.php	Manage the single GEN ED coordinator account
admin_requests.php	Review/approve conflict_requests
global_reports.php	Cross-college analytics and exports
faculty_teaching_load.php	Teaching load report (admin sees all colleges)
classroom_materials_monitor.php	Monitor classroom content uploads
dashboard.php	Admin dashboard statistics
settings.php	Admin profile settings

Shared Includes

File	Purpose
includes/admin_nav.php	render_admin_nav_sections() sidebar
includes/admin_offcanvas.php	Mobile off-canvas navigation
includes/admin_activity_log.php	Audit logging for admin actions
includes/mail_helpers.php	Credential emails for deans/GEN ED

admin_colleges.php

- Lists colleges with code, name, status (active/inactive).
 - Add/edit college records.
 - Cannot delete colleges with dependent data (FK constraints).
 - Logs changes via log_admin_activity().
-

admin_deans.php

- Creates dean users rows linked to colleges.dean_user_id.
 - Fields: username, full name, email, password (manual or generated).
 - Email temporary credentials with send_dean_credentials_mail().
 - Sets college_id on user and updates college dean reference.
-

admin_gened.php

- Manages the single institution-wide GEN ED account (role = 'gened').
- Update username, full name, email, active status.
- Password reset or generate-and-email temporary password.
- Requires upgrade_roles.php for users.email column.

admin_requests.php

- Lists pending conflict_requests from deans/chairs/faculty.
 - Approve → creates schedule via conflict override path.
 - Reject → sets status and admin_remarks.
 - Uses verify_admin_password() for sensitive approvals.
-

global_reports.php

- Institution-wide reporting beyond single-college dean scope.
 - Exports and summary charts for schedules, faculty, courses.
 - Access: require_role(['admin']).
-

Activity Logging (includes/admin_activity_log.php)

Function	Description
log_admin_activity(\$action, \$module, \$ref, \$before, \$after)	Primary audit entry point
log_user_activity(...)	Generic user activity
admin_activity_log_list_sorted(\$order)	Retrieve logs for display
admin_activity_log_collect_diff_lines(\$before, \$after)	Human-readable change diff

Stored in admin_activity_logs with JSON details_json snapshots.

Navigation (render_admin_nav_sections)

Section	Items
Overview	Dashboard, Messages
Institution	Colleges, Deans, GEN ED account
Scheduling	Schedules, Weekly view, Conflicts, Admin requests
Reports	Faculty teaching load, Global reports
Monitoring	Classroom materials monitor
Account	Settings

Related Tables

colleges, users, conflict_requests, schedules, admin_activity_logs, online_classrooms, classroom_content

Module 6: Dean (College)

College-scoped academic administration.

Role: dean

Scope: users.college_id → one college

Source Files

File	Purpose
dean_programs.php	Manage college_programs, year levels, sections
program_chairs.php	Assign program chair accounts to programs
faculty.php	Faculty CRUD for the college
courses.php	Course catalog (college-scoped, non-GE)
rooms.php	Room management for the college
add_schedule.php	Create new schedule entries
edit_schedule.php	Modify existing schedules
schedule.php	Schedule list view
view_schedule.php	Weekly calendar view
auto_schedule.php	Automated schedule generation
conflicts.php	View scheduling conflicts
dean_gened_chair.php	Assign GE program chair / CAS dean liaison
faculty_teaching_load.php	College teaching load report
reports.php	College reports
classroom_materials_monitor.php	Monitor uploaded materials
dashboard.php	Dean dashboard stats

Scope Enforcement

All dean pages use:

```
$collegeId = dean_college_id_or_fail();
```

SQL queries append college filter via `sql_scope_for_college()` or explicit `college_id = ?`.

Dean also sees applicable GE schedules via `dean_schedule_scope_sql()`.

dean_programs.php

- CRUD on college_programs (program code, name, status).
 - Manage program_year_levels and program_sections.
 - `parse_dean_program_year_levels_post()` processes form year-level checkboxes.
 - `program_year_levels_replace()` syncs year levels.
-

program_chairs.php

- Creates users with role = program_chair.
- Sets assigned_program and college_id.

- One chair per program (typical).
 - Credential email on create.
-

faculty.php

- CRUD `faculty` rows: `faculty_id`, `name`, `department`, `email`, `max hours`, `employment status`.
 - Link to users account (`user_id`) for login.
 - Manage `faculty_specializations` (courses faculty can teach).
 - Suggests usernames via `suggest_available_usernames()`.
-

courses.php

- College course catalog in `courses` where `is_gened = 0`.
 - Fields: `code`, `name`, `units`, `lecture/lab units`, `year level`, `section`, `department`.
 - `classroom_code` auto-allocated for LMS integration.
 - Shared with program chair (scoped to program department).
-

rooms.php

- College rooms (`is_gened = 0`, `college_id` set).
 - Auto room code generation via `next_auto_room_code_for_college()`.
 - Types: `lecture`, `laboratory`, `conference`, `tba`.
-

Scheduling Pages

Page	Dean capabilities
<code>add_schedule.php</code>	New offering with conflict check
<code>edit_schedule.php</code>	Update times, room, faculty
<code>schedule.php</code>	Filterable list
<code>view_schedule.php</code>	Weekly grid
<code>auto_schedule.php</code>	Batch auto-assign faculty/rooms
<code>conflicts.php</code>	Faculty/room/time conflicts

Conflict detection: `checkConflicts()` in `includes/functions.php`.

Schedule change requests from faculty appear on dean dashboard.

dean_gened_chair.php

- Configures GE liaison dean for messaging and live sessions.
 - Related to `GE_DEAN_USER_ID` / `GE_DEAN_NAME_HINT` config.
-

Activity Logging

log_dean_activity(\$actionType, \$details) → dean_activity_logs.

Navigation (dean sections in role_nav_sections)

Section	Items
Overview	Dashboard, Messages, Schedules, Weekly view
Academic	Faculty, Program chairs, Programs, Courses, Rooms
Scheduling	Add schedule, Auto schedule, Conflicts
Reports	Reports, Teaching load, Materials monitor
GE	GE chair assignment
Account	Settings

Related Tables

college_programs, program_year_levels, program_sections, faculty, courses, rooms, schedules, schedule_change_requests, dean_activity_logs

Module 7: Program Chair

Program-scoped management within a dean’s college.

Role: program_chair

Scope: users.college_id + users.assigned_program (department/program name)

Source Files

File	Purpose
faculty.php	Faculty in assigned program (department filter)
courses.php	Courses for assigned program
add_schedule.php	Create schedules for program offerings
edit_schedule.php	Edit program schedules
schedule.php	Schedule list (program-scoped)
view_schedule.php	Weekly view
conflicts.php	Conflict review
program_chair_students.php	Manage classroom_students in program
program_chair_student_registrations.php	Approve/reject registration requests
faculty_teaching_load.php	Teaching load for program faculty
reports.php	Program reports
classroom_materials_monitor.php	Content monitoring
dashboard.php	Chair dashboard (pending registrations count)

Scope Enforcement

```
$collegeId = dean_or_program_chair_college_id_or_fail();  
$programScope = program_scope_or_fail();
```

Queries filter: - faculty.department = \$programScope - courses.department = \$programScope - schedules joined to courses with matching department

program_chair_students.php

- Lists students (classroom_students) enrolled in program classrooms.
 - Create student accounts linked to users (role = student).
 - Manage student number, year level, email.
-

program_chair_student_registrations.php

Uses includes/student_registration_helpers.php:

Function	Description
pending_registrations_for_program(\$collegeId, \$programScope)	Pending queue
approve_student_registration_request(...)	Creates user + student row
reject_student_registration_request(...)	Sets rejection reason
count_pending_registrations_for_program(...)	Badge count for nav

Workflow: 1. Student submits registration on login.php. 2. Request stored in student_registration_requests (status = pending). 3. Chair approves → create_student_account() or rejects with reason.

Shared Pages with Dean

Program chairs share these pages with deans but see filtered data:

- faculty.php, courses.php, rooms.php (read-only rooms typically)
 - All scheduling pages
 - conflicts.php, reports.php
-

Navigation (program_chair in role_nav_sections)

Section	Items
Overview	Dashboard, Messages, Schedules, Weekly view
Academic	Faculty, Courses, Students, Registrations
Scheduling	Add schedule, Conflicts
Reports	Reports, Teaching load, Materials monitor
Account	Settings

Related Tables

faculty, courses, schedules, classroom_students, student_registration_requests, schedule_change_requests, classroom_enrollments

Module 8: Faculty

Instructor-facing tools for teaching, classrooms, and schedules.

Role: faculty

Identity: users linked to faculty via user_id or name match

Source Files

File	Purpose
faculty_schedule.php	Personal teaching schedule
schedule.php	Schedule list (faculty's classes)
view_schedule.php	Weekly calendar
conflicts.php	View conflicts affecting faculty
faculty_classrooms.php	List online classrooms for faculty
faculty_student_list.php	Courses with registered students and present/absent totals
faculty_classroom.php	Single classroom hub (content, students, settings)
faculty_classroom_week.php	Week-by-week content view
faculty_classroom_attendance.php	Attendance sessions and records
faculty_classroom_assessments.php	Quizzes, assignments, grading
faculty_edutools.php	EduTools launcher
faculty_teaching_load.php	Own teaching load summary
classroom_content_attachment.php	Download content attachments
classroom_syllabus.php	Upload/view syllabus PDF
messages.php	Internal messaging
reports.php	Faculty reports
settings.php	Profile settings
dashboard.php	Faculty dashboard

Identity Resolution

On login, `resolve_faculty_id_for_user()` sets `$_SESSION['faculty_id']`.

`faculty_college_id($facultyId)` scopes schedule queries.

faculty_classrooms.php

- Lists `online_classrooms` where `faculty_id` matches session.

- Each classroom links to a schedules row / courses row.
 - Shows join code, enrollment count, status.
-

faculty_student_list.php

- Lists the faculty member's online classrooms (courses) with registered student counts.
 - Present/absent totals come from classroom_attendance_records across class sessions.
 - Detail view (?id={classroom_id}) shows each enrolled student and their present/absent mark counts.
-

faculty_classroom.php

Central LMS management page:

Feature	Tables
Classroom settings	online_classrooms (title, meet link, grade weights)
Content (materials, links, announcements)	classroom_content
Student roster	classroom_enrollments + classroom_students
Discussion	classroom_messages via classroom_discussion_helpers.php
Join code	online_classrooms.join_code

faculty_classroom_attendance.php

- Creates classroom_attendance_sessions per class day.
 - Records classroom_attendance_records (present/absent).
 - Auto-attendance from online login (source = auto_login_online).
 - Manual override supported.
-

faculty_classroom_assessments.php

- CRUD classroom_assessments (written_work, performance_task, assignment, quiz).
 - Grade classroom_scores per student.
 - Review classroom_submissions for assignments.
 - Objective question auto-grading via classroom_grade_question_submission().
-

faculty_edutools.php

Launcher for faculty academic tools (links to classrooms, schedule, etc.).

Schedule Change Requests

Faculty can request schedule changes → schedule_change_requests reviewed by dean/chair.

Makeup classes

Faculty can also request a **makeup class** from **My schedule** (day, time, room, reason).
On approval, dean/chair automatically gets a temporary schedules row with `is_makeup = 1`.
Delete that makeup row after the class is held.

Navigation (faculty in `role_nav_sections`)

Section	Items
Overview	Dashboard, Messages, Schedules, Weekly view
Academic	Classrooms, Student list, My schedule, EduTools
Management	Reports, Teaching load, Conflicts
Account	Settings

Related Tables

`faculty`, `schedules`, `online_classrooms`, `classroom_*`, `schedule_change_requests`, `internal_messages`

Module 9: General Education (GEN ED)

Institution-wide General Education scheduling and resources.

Role: `gened`

Account: Single coordinator user (`role = gened`), managed via `admin_gened.php`

Source Files

File	Purpose
<code>gened_courses.php</code>	GE course catalog (<code>is_gened = 1</code>)
<code>gened_faculty.php</code>	GE faculty (<code>faculty.is_gened = 1</code>)
<code>gened_rooms.php</code>	GE rooms (<code>rooms.is_gened = 1</code>)
<code>gened_schedule.php</code>	GE schedule list
<code>gened_edit_schedule.php</code>	Add/edit GE schedule entries
<code>gened_assignments.php</code>	Map GE courses to colleges (<code>ge_course_colleges</code>)
<code>add_schedule.php</code> / <code>edit_schedule.php</code>	Shared schedule forms (GE mode)
<code>schedule.php</code>	Schedule list (GE scope)
<code>view_schedule.php</code>	Weekly view
<code>conflicts.php</code>	GE scheduling conflicts
<code>faculty_teaching_load.php</code>	GE faculty load
<code>classroom_materials_monitor.php</code>	Content monitoring
<code>reports.php</code>	GE reports
<code>messages.php</code>	Messaging (GE faculty ↔ GE dean rules)
<code>dashboard.php</code>	GEN ED dashboard

GE Data Model

Courses (courses where is_gened = 1)

- No college_id (institution-wide).
- Mapped to colleges via ge_course_colleges.

Faculty (faculty where is_gened = 1)

- college_id may be null.
- Teaches GE offerings only.

Rooms (rooms where is_gened = 1)

- room_code_scope = CONCAT('G|', room_code).
- next_auto_room_code_gened() for code generation.

Schedules

- Standard schedules row with GE course/faculty/room.
- Target audience stored in ge_schedule_targets:
 - college_id, program_name, year_level, section

gened_assignments.php

- Links GE courses to colleges that offer them.
- CRUD on ge_course_colleges.
- Drives which colleges see GE courses in dean schedules.

gened_edit_schedule.php

- GE-specific schedule form via render_schedule_form().
- Sets ge_schedule_targets for program/year/section targeting.
- Conflict checks include cross-college room sharing.

GE Live Sessions

weekly_schedule_gened_can_join_online_live() controls who can join GE online live classes: - GE faculty teaching the class - Target college deans/chairs - Configured GE dean (GE_DEAN_USER_ID)

Messaging Rules (messaging_helpers.php)

GE-specific messaging constraints: - messaging_is_gened_faculty_user() — identifies GE faculty - messaging_ge_dean_user_id() — resolves GE dean for messaging - GE faculty may only message the designated

Navigation (gened in role_nav_sections)

Section	Items
Overview	Dashboard, Messages, Schedules, Weekly view
GE Resources	GE courses, GE faculty, GE rooms, GE assignments
Scheduling	Add/edit GE schedule, Conflicts
Reports	Reports, Teaching load, Materials monitor
Account	Settings

Related Tables

courses, faculty, rooms, schedules, ge_course_colleges, ge_schedule_targets, faculty_specializations

Module 10: Student

Student portal for classrooms, schedules, and wellness.

Role: student
Identity: users linked to classroom_students via user_id

Source Files

File	Purpose
login.php	Student login + self-registration form
student_classrooms.php	List enrolled / joinable classrooms
student_classroom.php	Single classroom view (content, assessments)
student_classroom_week.php	Week-organized content view
student_edutools.php	EduTools hub (wellness, etc.)
student_wellness.php	Wellness check-in UI
classroom_content_attachment.php	Download attachments
view_schedule.php	Student schedule view (if applicable)
messages.php	Messaging with faculty
settings.php	Profile settings
dashboard.php	Student dashboard

Registration Flow

On login.php (mode=register):

1. Student fills: username, password, full name, email, student number, college, program, year level.
2. submit_student_registration() validates via validate_student_registration_input().

3. Inserts `student_registration_requests` (`status = pending`).
4. Program chair approves on `program_chair_student_registrations.php`.
5. `approve_student_registration_request()` creates users + `classroom_students`.

API support: `api/programs_by_college.php` — dynamic program/year-level dropdowns.

includes/student_registration_helpers.php

Function	Description
<code>student_registration_table_ready()</code>	Checks table exists
<code>active_colleges_for_registration()</code>	Active colleges list
<code>active_programs_for_college(\$collegeId)</code>	Programs for dropdown
<code>active_year_levels_for_program(...)</code>	Year levels for program
<code>submit_student_registration(...)</code>	Create pending request
<code>create_student_account(...)</code>	Create approved user + student
<code>username_taken_for_registration(\$username)</code>	Uniqueness check

student_classrooms.php

- Shows classrooms student is enrolled in (`classroom_enrollments`).
 - Join by code: `classroom_normalize_join_code()` + lookup `online_classrooms.join_code`.
 - Enrolls via `classroom_enrollments insert`.
-

student_classroom.php

Student view of classroom: - Read content (`classroom_content`) grouped by week - View announcements and links - Submit assessments → `classroom_submissions` - View scores → `classroom_scores` - Participate in discussion (`classroom_messages`)

student_edutools.php

Hub linking to: - Wellness check-in (`student_wellness.php`) - Other student academic tools

student_wellness.php

- Renders wellness companion UI via `includes/student_wellness_ui.php`.
 - Chat powered by `api/student_wellness_chat.php`.
 - See [14-wellness.md](#).
-

Messaging

Students can message faculty they are enrolled with: -
messaging_student_can_message_faculty(\$studentUserId, \$facultyUserId) - Enforced in
messaging_can_send()

Navigation (student in role_nav_sections)

Section	Items
Overview	Dashboard, Messages
Learning	Classrooms, EduTools
Account	Settings

Related Tables

classroom_students, student_registration_requests, classroom_enrollments, online_classrooms,
classroom_content, classroom_assessments, classroom_submissions, classroom_scores, internal_messages

Module 11: Scheduling

Core class scheduling engine shared across dean, program chair, GEN ED, faculty, and admin roles.

Source Files

File	Roles	Purpose
schedule.php	admin, dean, chair, gened, faculty	Filterable schedule list
view_schedule.php	admin, dean, chair, gened, faculty	Weekly grid calendar
add_schedule.php	dean, program_chair	Create schedule (college)
edit_schedule.php	dean, program_chair	Edit schedule (college)
gened_schedule.php	gened	GE schedule list
gened_edit_schedule.php	gened	Create/edit GE schedules
auto_schedule.php	dean	Automated assignment
conflicts.php	admin, dean, chair, gened, faculty	Conflict viewer
includes/schedule_helpers.php	—	Shared schedule form renderer
includes/functions.php	—	Conflict detection, validation rules

Data Model: schedules

Column	Description
faculty_id	Instructor
course_id	Course offering
room_id	Physical room
college_id	Owning college (null for some GE)
schedule_type	MW, TTH, MWF, Custom, etc.

Column	Description
day_of_week	MySQL SET of weekdays
start_time, end_time	Class period
semester, school_year, academic_year	Term identifiers
program, year_level, section	Audience
online_class_url, online_live_at	Virtual class links
created_by	User who created entry

Schedule Validation (`validate_schedule_rules`)

Enforces config constants:

Rule	Constant
Earliest start	TIME_MIN (06:00)
Latest end	TIME_MAX (22:00)
Minimum duration	MIN_CLASS_MINUTES (30)
Gap between classes same day	MIN_GAP_MINUTES
Max block duration	MAX_CLASS_BLOCK_HOURS
Max consecutive hours	MAX_CONSECUTIVE_HOURS

Conflict Detection (`checkConflicts`)

Checks before save:

1. **Faculty conflict** — Same faculty, overlapping days/times
2. **Room conflict** — Same room, overlapping times (unless room type = tba)
3. **Faculty gaps** — `check_faculty_gaps_and_consecutive()` for gap/consecutive rules
4. **Cross-college room** — `detect_cross_college_room_conflicts()` for shared rooms

Unresolved conflicts: - Logged to `conflict_logs` - Or submitted as `conflict_requests` for admin override

`render_schedule_form()` (`schedule_helpers.php`)

Shared HTML form used by add/edit pages:

Inputs: faculty, course, room, days, times, term, program, year, section, schedule type.

Options array controls: - College vs GE mode - Pre-filled defaults for edit - Available faculty/courses/rooms scoped to role

`auto_schedule.php`

Dean-only automated scheduler: - Assigns faculty and rooms to unscheduled course offerings - Respects faculty specializations, max hours, room capacity - Uses conflict detection iteratively

Weekly View (view_schedule.php)

- Renders 7-column week grid
 - Color-coded by course/faculty
 - Filters: semester, school year, college, program
 - Online live session indicators via weekly_schedule_online_live_mode()
-

GE Schedule Targets

GE schedules additionally write ge_schedule_targets: - Which college/program/year/section the GE class serves
- Used by deans to see GE classes in their college view

Schedule Change Workflow

Step	Actor	Table
1. Request	Faculty	schedule_change_requests
2. Review	Dean / Chair	Same table, status update
3. Apply	Dean / Chair	Update schedules row

Helper Functions Reference

Function	Purpose
fetch_faculty_day_schedules(\$facultyId, \$day, ...)	Faculty day schedule
fetch_room_day_schedules(\$roomId, \$day, ...)	Room day schedule
room_status_allows_overlap(\$roomId)	TBA rooms skip conflict
parse_day_set() / days_to_set()	Day SET conversion
create_conflict_request(\$payload)	Admin escalation

Related Tables

schedules, faculty, courses, rooms, conflict_logs, conflict_requests, schedule_change_requests, ge_schedule_targets, faculty_specializations

Module 12: Online Classroom (LMS)

Virtual classroom features tied to schedule entries.

Source Files

File	Role	Purpose
faculty_classrooms.php	faculty	Classroom list

File	Role	Purpose
faculty_classroom.php	faculty	Classroom management hub
faculty_classroom_week.php	faculty	Weekly content organizer
faculty_classroom_attendance.php	faculty	Attendance tracking
faculty_classroom_assessments.php	faculty	Assessments & grading
student_classrooms.php	student	Enrolled classrooms + join
student_classroom.php	student	Classroom learner view
student_classroom_week.php	student	Weekly content view
classroom_content_attachment.php	faculty, student	Secure file download
classroom_syllabus.php	faculty	Syllabus upload
classroom_materials_monitor.php	admin, dean, chair, gened	Upload monitoring
includes/classroom_discussion_helpers.php —		Discussion threads

Classroom Lifecycle

1. **Creation** — When a schedule is saved, an `online_classrooms` row may be created linking `schedule_id`, `faculty_id`, `course_id`.
 2. **Join code** — `classroom_alloc_unique_join_code()` assigns 8-char code.
 3. **Enrollment** — Students join via code → `classroom_enrollments`.
 4. **Content** — Faculty posts materials → `classroom_content`.
 5. **Assessment** — Faculty creates assessments → students submit → faculty grades.
-

Tables

`online_classrooms`

Field	Purpose
<code>schedule_id</code>	Links to physical schedule
<code>title</code> , <code>description</code>	Display info
<code>meet_link</code>	Video conference URL
<code>join_code</code>	Student enrollment code
<code>written_work_percentage</code> , <code>performance_task_percentage</code>	Grade weights
<code>syllabus_*</code>	Syllabus file metadata

`classroom_students` / `classroom_enrollments`

- Student roster independent of user accounts (can link via `user_id`).
- Many-to-many enrollment.

`classroom_content`

content_type	Use
<code>material</code>	Files, documents
<code>link</code>	External URLs

content_type	Use
announcement	Text announcements

Fields: title, body (rich HTML), weeks, days_per_topic, resource_url.

classroom_content_attachments

- Multiple files per content item.
- Stored under uploads/classroom/.

classroom_assessments

Types: written_work, performance_task, assignment, quiz.

classroom_submissions / classroom_scores

- Student answers and faculty grades.
- Unique per assessment + student.

classroom_attendance_sessions / classroom_attendance_records

- One session per classroom per day.
- Auto-present when student logs in during online window.
- Manual override by faculty.

classroom_messages

- Per-classroom discussion board.
- FIFO limit via classroom_discussion_enforce_fifo().

Content Helpers (includes/functions.php)

Function	Purpose
classroom_content_sanitize_html()	XSS-safe rich text
classroom_content_render_body()	Display sanitized HTML
classroom_content_group_by_week()	Group content by week label
classroom_content_store_attachment()	File upload handler
classroom_content_attachment_href()	Download URL builder
classroom_syllabus_href()	Syllabus download URL

Discussion Helpers (includes/classroom_discussion_helpers.php)

Function	Purpose
classroom_discussion_can_access()	Enrollment check
classroom_discussion_post()	New message
classroom_discussion_messages()	Fetch thread

Function	Purpose
<code>classroom_discussion_threads_for_user()</code>	Inbox across classrooms

Assessment Grading

Function	Purpose
<code>classroom_assessment_normalize_type()</code>	Valid type enum
<code>classroom_grade_question_submission()</code>	Auto-grade MC/true-false
<code>classroom_problem_answer_matches()</code>	Fuzzy answer matching

File Downloads

<code>classroom_content_attachment.php</code>	- Verifies user is faculty of classroom or enrolled student. - Streams file with correct Content-Type and download name. - Prevents directory traversal.
---	--

Materials Monitor

<code>classroom_materials_monitor.php</code>	- Admin/dean/chair/gened view of recent uploads. - Audit trail for institutional oversight.
--	---

Module 13: Messaging

Internal messaging between users with role-based permission rules.
--

Source Files

File	Purpose
<code>messages.php</code>	Main messaging UI (inbox, compose, thread view)
<code>message_attachment.php</code>	Download message attachments
<code>includes/messaging_helpers.php</code>	All messaging business logic

Roles with access: admin, dean, program_chair, gened, faculty, student

Data Model: internal_messages

Column	Description
<code>sender_user_id</code>	Author
<code>recipient_user_id</code>	Recipient
<code>subject</code>	Thread subject (memos)
<code>body</code>	Message text
<code>is_memo</code>	Broadcast memo flag

Column	Description
attachment_*	Optional file attachment
created_at	Timestamp
read_at	Read receipt (null = unread)

Key Functions (includes/messaging_helpers.php)

Infrastructure

Function	Description
messaging_table_exists()	Schema check
messaging_has_memo_columns()	Memo feature check
messaging_thread_max_messages()	FIFO limit from config
messaging_enforce_thread_fifo()	Delete oldest when over limit

Permissions

Function	Description
messaging_can_open_thread(\$viewerId, \$otherId)	Can view conversation
messaging_can_send(\$senderId, \$recipientId)	Can send message
messaging_allowed_recipients(\$forUserId)	Dropdown recipient list
messaging_faculty_can_message_student(...)	Enrollment-based rule
messaging_student_can_message_faculty(...)	Reverse enrollment rule
messaging_is_gened_faculty_user()	GE faculty detection
messaging_ge_dean_user_id()	GE dean for GE faculty

Operations

Function	Description
messaging_send(\$fromId, \$toId, \$body, \$options)	Send direct message
messaging_send_memo(\$fromId, \$recipientIds, ...)	Broadcast memo
messaging_thread(\$userId, \$otherId)	Fetch conversation
messaging_conversation_list(\$userId)	Inbox with previews
messaging_unread_count(\$userId)	Badge count for nav
messaging_mark_read(\$viewerId, \$otherId)	Mark thread read
messaging_delete_message(\$messageId, \$userId)	Delete own message

Attachments

Function	Description
messaging_store_attachment(\$file)	Save to uploads

Function	Description
<code>messaging_attachment_path(\$storedName)</code>	Filesystem path
<code>messaging_delete_attachment_if_unused()</code>	Cleanup orphaned files

Permission Matrix (Summary)

Sender → Recipient	Allowed when
Faculty → Student	Student enrolled in faculty's classroom
Student → Faculty	Same enrollment relationship
Dean → Program chairs	Same college
Program chair → Dean	Same college
GE faculty → GE dean	Configured GE dean only
Admin → Various	Broad access per <code>messaging_allowed_recipients()</code>
Memo broadcast	Dean/chair/admin to scoped recipients

FIFO Thread Limit

`MESSAGING_THREAD_MAX_MESSAGES` (default 10): - When exceeded, oldest messages in a thread are deleted. - `messaging_enforce_thread_fifo()` runs after each send.

UI (`messages.php`)

Features: - Conversation list with unread badges - Thread view with role badges (`messaging_role_badge_class()`) - Compose to allowed recipients - Memo mode for broadcast messages - File attachment upload - Nav badge via `messaging_unread_count()` in `admin_nav.php`

Attachment Download

`message_attachment.php`: - Verifies sender or recipient access. - Streams attachment with original filename.

Config

Constant	Purpose
<code>MESSAGING_THREAD_MAX_MESSAGES</code>	Per-thread message cap
<code>GE_DEAN_USER_ID</code>	GE dean user for GE faculty messaging
<code>GE_DEAN_NAME_HINT</code>	Fallback dean name matching

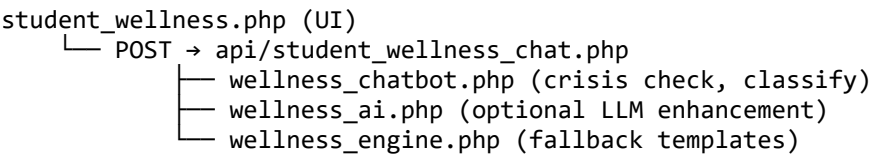
Module 14: Student Wellness

Non-diagnostic wellness companion with crisis escalation (Philippines resources).

Source Files

File	Purpose
student_wellness.php	Wellness check-in page (student UI)
student_edutools.php	Entry point from EduTools
api/student_wellness_chat.php	REST chat endpoint
api/wellness_chatbot.openapi.yaml	OpenAPI specification
includes/student_wellness_ui.php	HTML/JS chat widget
includes/wellness_chatbot.php	Classification, crisis detection, orchestration
includes/wellness_engine.php	Template-based empathetic replies
includes/wellness_ai.php	Groq/OpenAI/Gemini/Ollama integration

Architecture



API (api/student_wellness_chat.php)

Method	Auth	Description
GET	Public	Service descriptor / health
POST	Student session	Chat completion

POST body:

```
{
  "message": "user text",
  "lang": "en|tl|optional",
  "history": [{"role":"user|assistant", "content":"..."}]
}
```

Response:

```
{
  "reply": "assistant text",
  "crisis": false,
  "lang": "en",
  "disclaimer": "...",
  "provider": "builtin|groq|ollama"
}
```

Crisis Detection (wellness_chatbot.php)

Function	Purpose
wellness_is_crisis_message(\$norm)	Detects self-harm/suicide keywords

Function	Purpose
<code>wellness_crisis_response(\$lang)</code>	PH hotline resources (NCMH, Hopeline, etc.)
<code>wellness_ph_crises_resources_public()</code>	Public crisis resource list

Crisis messages bypass normal chat — immediate escalation response with hotlines.

Message Classification

Function	Purpose
<code>wellness_normalize_for_match(\$text)</code>	Lowercase, strip punctuation
<code>wellness_classify(\$norm)</code>	Topic + scenario scores
<code>wellness_score_scenarios(\$norm)</code>	stress, anxiety, sleep, academics, etc.
<code>wellness_chat_orchestrate(\$message, \$langHint, \$history)</code>	Main entry point

Template Engine (`wellness_engine.php`)

Fallback when AI is disabled:

Function	Purpose
<code>wellness_parse_student_message()</code>	Extract intent/context
<code>wellness_empathy_opening()</code>	Empathetic opener
<code>wellness_core_steps()</code>	Actionable coping steps
<code>wellness_soft_close()</code>	Closing encouragement
<code>wellness_engine_reply()</code>	Compose full reply

Supports English and Tagalog (tl).

AI Provider (`wellness_ai.php`)

Function	Purpose
<code>wellness_ai_is_enabled()</code>	Checks config
<code>wellness_ai_provider()</code>	groq, ollama, openai, gemini, auto
<code>wellness_ai_chat_completion(...)</code>	LLM API call

Providers: - **Groq** (default cloud) — `WELLNESS_AI_API_KEY` starting with `gsk_` - **Ollama** (local) — `WELLNESS_OLLAMA_URL`, `WELLNESS_OLLAMA_MODEL` - **builtin** — Template engine only

Configuration

Constant	Default	Purpose
<code>WELLNESS_AI_ENABLED</code>	<code>false</code>	Master switch
<code>WELLNESS_AI_PROVIDER</code>	<code>groq</code>	Provider selection

Constant	Default	Purpose
WELLNESS_AI_API_KEY	env GROQ_API_KEY	API key
WELLNESS_AI_MODEL	llama-3.3-70b-versatile	Model name
WELLNESS_OLLAMA_URL	http://127.0.0.1:11434	Local LLM

Disclaimers

All responses include non-diagnostic disclaimers: - wellness_disclaimer_banner(\$lang) - wellness_footer_disclaimer(\$lang)

Not a substitute for professional mental health care.

OpenAPI

api/wellness_chatbot.openapi.yaml documents the REST contract for external integration or testing.

Access Control

- Page: require_role(['student'])
- API POST: Validates \$_SESSION['role'] === 'student'

Module 15: Reports & Analytics

Reporting and monitoring across roles.

Source Files

File	Roles	Purpose
reports.php	admin, dean, chair, gened, faculty	Role-scoped reports hub
global_reports.php	admin	Institution-wide analytics
faculty_teaching_load.php	admin, dean, chair, gened	Teaching load in units
super_admin_teaching_load_report.php	super_admin	Cross-college load report
classroom_materials_monitor.php	admin, dean, chair, gened	Upload activity monitor
dashboard.php	all roles	Summary statistics widgets
dashboard1.php	—	Alternate dashboard layout

reports.php

General reports page scoped by role:

Role	Scope
admin	All colleges
dean	Own college
program_chair	Own program
gened	GE offerings
faculty	Own classes

Typical report types: - Schedule summaries by term - Faculty assignment counts - Course offering statistics - Export to printable views (`assets/css/print.css`)

global_reports.php

Admin-only cross-college analytics: - Aggregate schedule counts - College comparison charts - Term-over-term summaries - Institution KPIs

Teaching Load Reports

faculty_teaching_load.php

Calculates load from schedules JOIN courses:

`load_units = SUM(courses.units) per faculty per term`

Filters: - College (admin sees all; dean/chair scoped) - Semester, school year - Department/program - GE mode for gened role

super_admin_teaching_load_report.php

Super admin variant with institution-wide filters and export.

classroom_materials_monitor.php

Monitors `classroom_content` and attachments: - Recent uploads across classrooms - Filter by college, faculty, date range - Oversight for academic integrity / policy compliance

Dashboard Statistics (dashboard.php)

Role-specific stat cards:

Admin

- Active faculty (college-assigned)
- Total courses, schedules
- Open conflict requests

Dean / Program Chair

- Scoped faculty, courses, schedules
- Pending schedule change requests
- Pending student registrations (chair)

Faculty

- Today's classes
- Classroom count
- Upcoming assessments

Student

- Enrolled classrooms
- Pending assessments

Super Admin

- Extended KPIs (see [04-super-admin.md](#))

Auto-refresh: 30 seconds for admin dashboard.

Print Styles

assets/css/print.css — Optimized layouts for printing schedules and reports from browser.

Activity Logs (Reporting Sources)

Table	Used for
admin_activity_logs	Admin action audit
dean_activity_logs	Dean/chair action history
conflict_logs	Scheduling conflict history

Admin activity log viewer integrated in admin settings/offcanvas.

Related Helpers

Function	Module	Purpose
sql_scope_for_college()	functions.php	College SQL filter
dean_schedule_scope_sql()	functions.php	Dean + GE schedule scope
admin_activity_log_list_sorted()	admin_activity_log.php	Audit log retrieval

Module 16: REST API

JSON endpoints for AJAX and external clients.

Source Files

File	Method	Auth	Purpose
api/student_wellness_chat.php	GET, POST	Student (POST)	Wellness chat companion
api/programs_by_college.php	GET	Public	Programs/year-levels for registration
api/wellness_chatbot.openapi.yaml	—	—	OpenAPI 3 spec for wellness API

api/programs_by_college.php

Purpose: Powers dynamic dropdowns on student registration form.

Request:

GET /api/programs_by_college.php?college_id=1

Response:

```
{
  "programs": [
    {"id": 1, "program_name": "BSIT", "status": "active"}
  ],
  "year_levels_by_program": {
    "BSIT": ["1st Year", "2nd Year", "3rd Year", "4th Year"]
  }
}
```

Implementation: - Includes student_registration_helpers.php - Calls active_programs_for_college() and active_year_levels_by_program_for_college() - No session required (public for registration UX)

api/student_wellness_chat.php

See [14-wellness.md](#) for full documentation.

Summary:

Endpoint	Description
GET	Returns API metadata, disclaimer, crisis resources
POST	Processes chat message, returns AI/template reply

Headers: - Content-Type: application/json - X-Content-Type-Options: nosniff

Session: Uses SESSION_NAME cookie; POST requires role = student.

Error responses: - 401 — Not authenticated as student - 400 — Invalid JSON or empty message - 500 — Internal error

OpenAPI Spec

api/wellness_chatbot.openapi.yaml defines:

- /api/student_wellness_chat paths
- Request/response schemas
- Crisis response format
- Example payloads

Use with Swagger UI or code generators for client SDKs.

API Design Patterns

1. **Bootstrap:** Each endpoint includes config/config.php directly (not full page auth stack).
 2. **JSON only:** All responses Content-Type: application/json; charset=utf-8.
 3. **Session auth:** Wellness API uses PHP sessions, not API keys.
 4. **Error handling:** wellness_api_emit_json(\$code, \$body) helper with JSON_THROW_ON_ERROR.
-

Future Extension Points

The api/ directory is the designated location for new JSON endpoints. Follow existing patterns: - declare(strict_types=1) - Direct config include - Explicit auth check before business logic - JSON encode with JSON_UNESCAPED_UNICODE

Module 17: UI Assets & Shared Layout

Frontend resources and shared presentation layer.

Source Files

Stylesheets (assets/css/)

File	Purpose
style.css	Main application theme, layout, components
print.css	Print-optimized styles for schedules/reports

JavaScript (assets/js/)

File	Purpose
app_nav.js	Sidebar navigation behavior, active states
app_tooltips.js	Bootstrap tooltip initialization
theme-toggle.js	Light/dark theme switching

Layout Partial (includes/)

File	Purpose
header.php	<!DOCTYPE html>, meta, CSS links, opens <body>, top bar
footer.php	Closes containers, loads JS, </body></html>
admin_nav.php	Role-based sidebar navigation
admin_offcanvas.php	Mobile off-canvas menu wrapper
student_tooltip.php	app_tooltip_attr() for accessible tooltips

includes/header.php

Responsibilities: - Sets \$pageTitle in browser tab - Loads Bootstrap 5, Font Awesome icons - Loads assets/css/style.css - Renders top navigation bar with user avatar - Opens main content container - Includes role navigation via render_role_nav_sections()

includes/admin_nav.php

Navigation rendering functions:

Function	Role
render_super_admin_nav_sections()	super_admin sidebar
render_admin_nav_sections()	admin sidebar
role_nav_sections(\$role, \$unread)	Returns nav structure array
render_role_nav_sections()	Dispatches to correct renderer
render_nav_sections_markup()	HTML output with active highlighting
app_sidebar_brand_meta(\$role)	Brand subtitle per role
app_render_topbar_avatar()	Profile photo in top bar

Nav items include: - file — PHP filename for active state matching - href — Link target - icon — Font Awesome class - label — Display text - badge — Optional unread count (messages) - tooltip — Help text via app_tooltip_attr()

includes/admin_offcanvas.php

Mobile-responsive off-canvas sidebar: - Bootstrap offcanvas component - Duplicates nav for small screens - \$dismissOffcanvas flag closes menu on link click

Tooltips (includes/student_tooltip.php)

Function	Description
app_cursor_tooltips_active()	Feature flag check
app_tooltip_attr(\$text)	Returns data-bs-toggle="tooltip" attributes
student_tooltip_attr(\$text)	Alias for student-facing pages

Initialized by app_tooltips.js on DOM ready.

Theme Toggle (assets/js/theme-toggle.js)

- Persists preference in localStorage
 - Toggles data-bs-theme on <html>
 - Respects system preference on first visit
-

Navigation JS (assets/js/app_nav.js)

- Highlights active nav item based on current page filename
 - Collapsible sidebar sections
 - Mobile menu interactions
-

Common UI Patterns

Flash messages

```
$flash = $_SESSION['flash'] ?? '';
unset($_SESSION['flash']);
// ...
<?php if ($flash): ?><div class="alert alert-info"><?= htmlspecialchars($flash) ?></div><?php endif;
?>
```

Cards

Bootstrap card shadow-sm wrappers for forms and data tables.

Forms

Bootstrap 5 grid (row g-3, col-md-*), form-control, form-check.

Icons

Font Awesome 6 solid (fa-solid fa-*) throughout navigation and headings.

Page titles

```
$pageTitle = 'Page Name';
require_once __DIR__ . '/includes/header.php';
?>
<h1 class="h3 mb-4"><i class="fa-solid fa-icon me-2 text-primary"></i>Title</h1>
```

Upload Directories

Path	Contents
uploads/profile/	User profile photos
uploads/classroom/	Classroom content attachments
uploads/messages/	Message attachments

Path	Contents
uploads/syllabus/	Classroom syllabi

Protected by uploads/web.config (no script execution).

External Dependencies (CDN)

Loaded in header.php: - Bootstrap 5 CSS/JS - Font Awesome 6 - No npm/build step — plain static assets

settings.php

Shared profile page for all roles: - Update full name, email (if column exists) - Change password - Upload profile photo via profile_photo.php - Role-appropriate redirect after save

profile_photo.php

POST endpoint for avatar upload/remove: - Uses profile_photo_store() / profile_photo_remove() - Returns to settings with flash message

Appendix A: Key Source Code Excerpts

Representative source files referenced throughout this documentation.

index.php — Application entry point

```
<?php
declare(strict_types=1);

require_once __DIR__ . '/includes/auth.php';

if (!empty($_SESSION['user_id'])) {
    header('Location: dashboard.php');
} else {
    header('Location: login.php');
}
exit;
```

includes/db.php — Database connection

```
<?php
declare(strict_types=1);

require_once dirname(__DIR__) . '/config/config.php';

function db(): PDO
{
    static $pdo = null;
    if ($pdo === null) {
```

```

    $dsn = sprintf(
        'mysql:host=%s;port=%d;dbname=%s;charset=%s',
        DB_HOST,
        DB_PORT,
        DB_NAME,
        DB_CHARSET
    );
    try {
        $pdo = new PDO($dsn, DB_USER, DB_PASS, [
            PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
            PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
        ]);
    } catch (PDOException $e) {
        // ... XAMPP-friendly connection error ...
        throw $e;
    }
}
return $pdo;
}

```

includes/auth.php — Role gate

```

function require_role(array $roles): void
{
    require_login();
    if (!in_array((string) ($_SESSION['role'] ?? ''), $roles, true)) {
        http_response_code(403);
        echo 'Forbidden';
        exit;
    }
}

```

admin_gened.php — Admin page pattern

```

<?php
declare(strict_types=1);

require_once __DIR__ . '/includes/auth.php';
require_once __DIR__ . '/includes/functions.php';
require_once __DIR__ . '/includes/mail_helpers.php';
require_once __DIR__ . '/includes/admin_activity_log.php';

require_role(['admin']);
// POST handler → flash message → redirect
// GET → Load data → header.php → HTML form → footer.php

```

api/programs_by_college.php — JSON API pattern

```

<?php
declare(strict_types=1);

require_once __DIR__ . '/../config/config.php';
require_once __DIR__ . '/../includes/db.php';
require_once __DIR__ . '/../includes/student_registration_helpers.php';

header('Content-Type: application/json; charset=utf-8');

```

```
$collegeId = (int) ($_GET['college_id'] ?? 0);
$programs = active_programs_for_college($collegeId);
$yearLevelsByProgram = active_year_levels_by_program_for_college($collegeId);

echo json_encode([
    'programs' => $programs,
    'year_levels_by_program' => $yearLevelsByProgram,
], JSON_UNESCAPED_UNICODE);
```

install/schema.sql — Core tables (excerpt)

```
CREATE TABLE IF NOT EXISTS users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50) NOT NULL UNIQUE,
    password VARCHAR(255) NOT NULL,
    full_name VARCHAR(100) NOT NULL,
    role ENUM('super_admin', 'admin', 'dean', 'program_chair', 'faculty', 'gened', 'student') NOT NULL,
    college_id INT NULL,
    is_active TINYINT(1) NOT NULL DEFAULT 1,
    ...
);

CREATE TABLE IF NOT EXISTS schedules (
    id INT AUTO_INCREMENT PRIMARY KEY,
    faculty_id INT NOT NULL,
    course_id INT NOT NULL,
    room_id INT NOT NULL,
    day_of_week SET('Monday', 'Tuesday', ...) NOT NULL,
    start_time TIME NOT NULL,
    end_time TIME NOT NULL,
    semester VARCHAR(20) NOT NULL,
    school_year VARCHAR(20) NOT NULL,
    ...
);
```